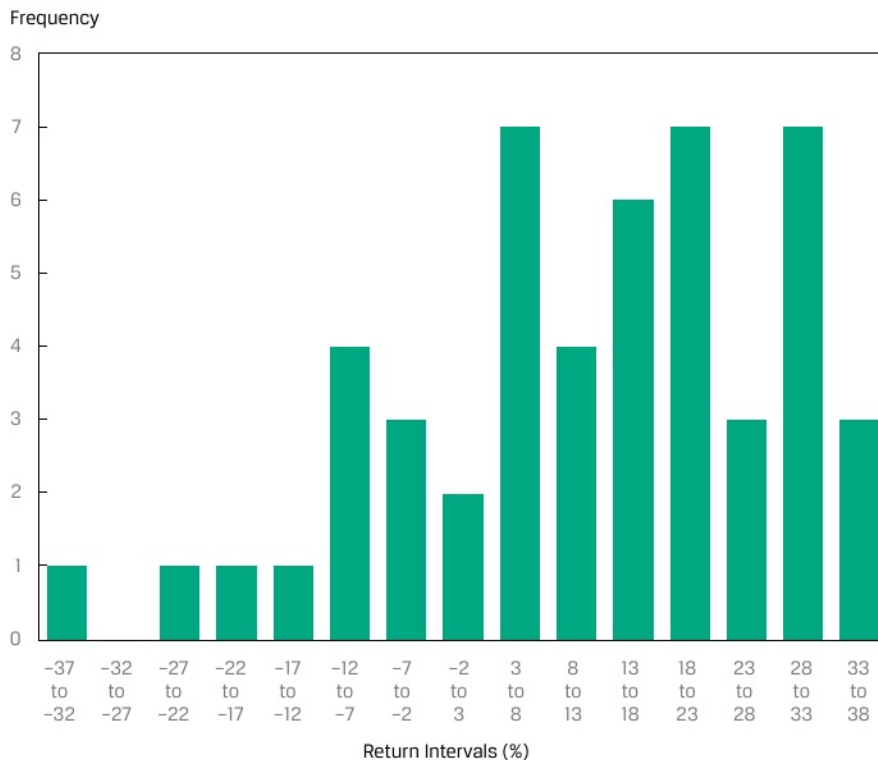


QUESTION SET



The histogram in Exhibit 7 shows a distribution of the annual returns for the S&P 500 Index for a 50-year period.

Exhibit 7: Annual Returns for the S&P 50 Index



1. The bin containing the median return is:

- A. 3 percent to 8 percent.
- B. 8 percent to 13 percent.
- C. 13 percent to 18 percent.

Solution:

C is correct. Because 50 data points are in the histogram, the median return would be the mean of the $50/2 = 25$ th and $(50 + 2)/2 = 26$ th positions. The sum of the return bin frequencies to the left of the 13 percent to 18 percent interval is 24. As a result, the 25th and 26th returns will fall in the 13 percent to 18 percent interval.

2. Based on Exhibit 7, the distribution would be *best* described as being:

- A. unimodal.
- B. bimodal.
- C. trimodal.

Solution:

C is correct. The mode of a distribution with data grouped in intervals is the interval with the highest frequency. The three intervals of 3 percent to 8